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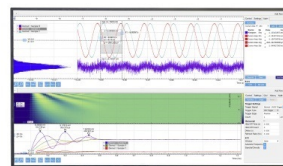
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Flexible transition state theory for a variable reaction coordinate: Derivation of canonical and microcanonical forms

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A completely general canonical and microcanonical (energy-resolved) flexible transition state theory (FTST) expression for the rate constant is derived for an arbitrary choice of reaction coordinate. The derivation is thorough and rigorous within the framework of FTST and replaces our previous treatments [Robertson *et al.*, *J. Chem. Phys.* **103**, 2917 (1995); Robertson *et al.*, *Faraday Discuss. Chem. Soc.* **102**, 65 (1995)] which implicitly involved some significant assumptions. The canonical rate expressions obtained here agree with our earlier results. The corresponding microcanonical results are new. The rate expressions apply to any definition of the separation distance between fragments in a barrierless recombination (or dissociation) that is held fixed during hindered rotations at the transition state, and to any combination of fragment structure (atom, linear top, nonlinear top). The minimization of the rate constant with respect to this definition can be regarded as optimizing the reaction coordinate within a canonical or microcanonical framework. The expression is analytic except for a configuration integral whose evaluation generally requires numerical integration over internal angles (from one to five depending on the fragment structures). The form of the integrand in this integral has important conceptual and computational implications. The primary component of the integrand is the determinant of the inverse *G*-matrix associated with the external rotations and the relative internal motion of the fragments. © 2000 American Institute of Physics. [S0021-9606(00)00531-6]

I. INTRODUCTION

Reactions whose transition states or “bottlenecks” are characterized by large amplitude motion constitute a large and important class of chemical processes. This situation most commonly arises for recombination of molecular radicals, or the reverse process of single-bond fission, for which there is no (pronounced) potential barrier for the formation of a parent molecule from a pair of constituent molecular fragments. Such “barrierless” reactions are commonplace, particularly in combustion, atmospheric, and interstellar processes. Despite ongoing advances in the implementation of classical trajectories and quantum dynamics, statistical theories remain the most widely used approach for modeling reaction rate coefficients. The accurate incorporation of large amplitude motion into statistical theories of reaction kinetics is essential if such theories are to be relied upon for predictions, for interpretation of measurements, and for development of chemical intuition about “floppy” transition states. This endeavor has been, and continues to be, the subject of substantial theoretical attention, e.g., Refs. 1–4.

Here we present further developments for accurately incorporating such large amplitude motion within a variational transition state theory (VTST) framework. This paper will deal exclusively with flexible transition state theory (FTST), a version of VTST designed specifically to handle large am-

plitude motion.⁵ Like all modern theories for barrierless reactions, FTST is a variational theory, meaning that rate constant is minimized by varying the transition state location along the reaction coordinate. This minimization is alternatively referred to as “optimization” of the location of the bottleneck in the reactive flux. Another approach for dealing with large amplitude motion at the transition state is provided by adiabatic channel theory. The most widely applied model of this type is the statistical adiabatic channel model (SACM)^{3,6} which incorporates a quantal treatment of all modes at the expense of an approximate and unspecified description of the transitional mode potential. The kinetics implications of the approximations in FTST versus those in SACM (and other theories) remains an issue for further study.

A potentially important issue in applications of FTST is the choice of reaction coordinate (RC). In statistical theories of reaction dynamics, such as FTST, a dividing surface (DS) orthogonal to the RC is introduced and the critical DS is the one which minimizes the statistical (canonical or microcanonical) reactive flux. The statistical flux through a DS is directly proportional to the number of accessible states on the DS. Different DS's are explored by varying both the definition of the RC and the RC value for a given definition. The variationally optimized DS corresponds to the transition state (TS). In original FTST⁷ the reaction coordinate was

restricted to be the separation of the centers-of-mass of the fragments, R_{cm} . A centers-of-mass reaction coordinate (CMRC) provides a natural and simple description of the kinetic energy of the fragments and is certainly appropriate for well-separated fragments where the inter-fragment potential has effectively vanished. At intermediate and small separations, where the potential is significant and may be highly anisotropic, the validity of a CMRC is suspect. Extensive work by Klippenstein^{8,9} in the early 90's within a microcanonical FTST framework did indeed reveal the limitations of a CMRC for several representative systems. Klippenstein has shown that use of a bond reaction coordinate (BRC), i.e., atom-atom separation distance, can dramatically lower the sum of states at intermediate and small separations. He also introduced the concept of a generalized RC and implemented this idea for a limited set of reaction coordinate definitions for dissociations involving linear (diatomic) fragments.⁸⁻¹⁰

This paper is the third in a series by the present authors setting out a systematic reformulation and extension of FTST. The objectives of the series are (i) streamlined numerical implementation (without loss of rigor) compared to original FTST⁷ which involves intensive numerical calculations of phase space integrals, (ii) improved physical insight into effect of the transitional mode potential energy on the rate constant, (iii) incorporation of a generalized reaction coordinate to extend the range of applicability, (iv) derivation of analytic FTST rate constant expressions for a generalized reaction coordinate, (v) understanding the physical significance of variationally optimized reaction coordinates, and (vi) the derivation of symmetry-based restrictions on the variations in a generalized reaction coordinate. The first paper¹¹ in the series (hereafter denoted paper I) addressed objectives (i) and (ii). The second paper¹² (hereafter paper II) began to address objective (iii) by setting out the structure of the required formalism; a companion paper (hereafter paper IV) deals with (iv) and begins to address (v). A forthcoming paper (in preparation) will address (vi).

In paper II, we extended our reformulated version of *canonical* FTST (CFTST) with a CMRC (as obtained in paper I) to the case of an arbitrary RC. We obtained in paper II a general expression for the CFTST rate constant with a variable reaction coordinate (VRC), whose form and numerical implementation involve relatively minor modifications of the CMRC treatment described in paper I. In paper II we also derived the specific analytic CFTST-VRC rate expression for the case of atom+linear top fragments. Specific rate expressions for the other four cases (atom-nonlinear top, linear top-linear top, linear-nonlinear, nonlinear-nonlinear) were subsequently obtained and applied, but have not been published.

The current paper subsumes the formal development of paper II for the incorporation of a VRC into FTST. The treatment in paper II is incomplete in the sense that it does not start from "first principles" with the classical flux integral over all relevant coordinates and momenta, including those for the RC. Instead, paper II begins after integration over the RC and its conjugate momentum have been (implicitly) carried out, leading to ambiguity as to the significance and validity of the ensuing results. The current treatment

puts the results of paper II on a more rigorous foundation. In addition to a complete treatment, it provides a more transparent derivation based on matrices, thus avoiding the algebraic treatment involving Cartesian coordinates employed in paper II. Also, in the current paper we extend our reformulated version of FTST to the microcanonical (E-resolved) level. The companion paper completes the detailed implementation begun in paper II, by providing explicit FTST-VRC rate constant expressions for all five combinations of fragment types.

The paper is organized as follows; Section II will review FTST. In Sec. III the CFTST rate expression for a center-of-mass reaction coordinate is rederived, beginning with the fundamental definition of the statistical rate constant in terms flux through a dividing surface. This derivation also serves to introduce notation and concepts which will be used in the next section. In Sec. IV, the corresponding canonical rate expression for an arbitrary reaction coordinate is rigorously derived, and in a manner which clearly relates it to its CM counterpart derived in Sec. III. In Sec. V the energy-resolved microcanonical FTST rate expression for an arbitrary reaction coordinate is obtained by Laplace transformation of the canonical result. Section VI summarizes the paper.

II. FTST BACKGROUND

A. Assumptions

In FTST, the degrees of freedom at the transition state are separated into two groups, called the transitional modes and the conserved modes, for the purposes of state counting. The transitional modes involve large amplitude relative motion of the fragments and are so named because they correspond to free rotations of the separated reactants that evolve into vibrational motions of the parent molecule. The conserved degrees of freedom are those modes which do not change appreciably in character while going from reactants to products. These modes are treated quantumly using standard state counting techniques.¹³ They may be modeled as independent harmonic or anharmonic oscillators; approximate correction factors for inter-mode coupling may be applied if appropriate. Their relatively high frequencies make the effects of zero point energy and a discrete state density important. The enumeration of modes of the conserved type is a well-developed field to which no further attention is given here.

The cornerstone of FTST is an accurate collective treatment of the transitional modes. Full coupling is retained in the form of a classical Hamiltonian, $H_{\text{tr}} = T + V_{\text{tr}}$, where V_{tr} is the full transitional mode potential for relative motion of the fragments and T is the full kinetic energy, including coupling of these modes with overall rotation. In describing T and V_{tr} the fragments are assumed to be rigid bodies whose shapes at the given value R of the reaction coordinate are determined by the minimum energy path (MEP) configuration associated with this value of R . That is, at any point on the MEP, each fragment has an optimized internal geometry in response to the presence of the other fragment. When either fragment departs from the MEP in transitional mode directions such that R is fixed, its internal geometry will be

presumed fixed and only its orientation relative to the other fragment will change. This assumption means that each fragment, no matter how complicated, can be treated as either an atom or a linear, spherical, symmetric, or asymmetric top for the purposes of transitional motion. Principal moment of inertia analysis of each fragment at position R along the MEP fixes the locations of the atoms of the fragment with respect to the principal axes. The relative orientation of the fragments is then determined by only the angles that govern the relative orientation of the principal axes of each top. When segregated from the full potential energy surface (PES), V_{tr} can ultimately be formulated as a function of only those orientation angles and R .

The accurate classical treatment of the transitional modes is made possible by assuming a strict separation from the conserved modes. Such a separation is physically reasonable on the basis of different time scales for the two types of motion. In the region of flexible transition states, transitional modes are typically characterized by low frequency motions, whereas the conserved modes typically consist of relatively high frequency stretching and bending vibrations. The fact that conserved mode frequencies generally depend weakly on fragment separation further suggests that these modes are not strongly coupled to the transitional modes. In the vicinity of the reaction path (RP) (defined as the path of steepest descent in mass weighted coordinates on the full PES), the extent of this coupling can be examined by performing an RP analysis of the Hamiltonian.¹⁴ Such analyses are, however, tedious, and have been performed for a limited number of systems. For $\text{CH}_3 + \text{H}$, a barrierless recombination reaction, such an analysis has been done,¹⁵ revealing small coupling coefficients between transitional modes and conserved modes of the same symmetry (otherwise the coupling is identically zero). Also neglected in FTST are the quantum effects of tunneling and transitional mode zero point energy. These have been investigated and found to be insignificant, at least for the selected test cases. At the temperature-resolved (canonical) level, Klippenstein and Marcus¹⁶ showed that quantum and classical treatments of the transitional modes in $\text{CH}_3 + \text{CH}_3$ resulted in rate constants differing by no more than 3% over a wide temperature range. At the energy resolved (microcanonical) level, separate studies of $\text{CH}_3 + \text{H}$ by Pilling and co-workers;¹⁷ and by Aubanel and Wardlaw,¹⁸ showed that the effects of reaction coordinate tunneling and transitional mode zero point energy, respectively, are negligible.

Another motivation for separating the transitional and conserved modes is that complete PESs (for all internal degrees of freedom) exist for only a few barrierless reactions, a situation unlikely to change in the near future. Furthermore, even if a complete PES exists, a full quantum state count including all couplings and anharmonicities for all degrees of freedom is not currently feasible. The assumptions in FTST, namely separation of transitional and conserved modes and classical treatment of the transitional modes, appear to be optimal among possible approximate treatments. Compelling evidence for the reliability of FTST is provided by comparison to experiments: Klippenstein has documented cases of

detailed agreement between measurements and FTST calculations at the microcanonical level.²

B. Degrees of freedom

There are different ways of allocating degrees of freedom to a system of two associating fragments. To make the discussion concrete we focus on the case of two nonlinear tops—the allocation for other systems is straightforward. One of the simplest approaches is to treat each fragment independently, assigning to each three Cartesian coordinates specifying the position of the fragment's center-of-mass (CM) in a space-fixed system, and three Euler angles specifying orientation of a fragment-fixed Cartesian system with origin at the fragment CM relative to the space-fixed system. The remaining degrees of freedom refer to the internal motions of the fragments and are deemed to be conserved modes. It is difficult to proceed with this assignment because, among other things, the coordinate that will serve as the reaction coordinate must in general be expressed as a function of these 12 coordinates.

Klippenstein and Marcus¹⁶ adopted a coordinate system related to the above in which they retained the Euler angle specification of the orientation of the each fragment relative to space fixed axes and the identification of the internal modes of the fragments with the conserved modes. However, instead of assigning three coordinates to the individual fragment CMs they assigned three coordinate to the overall CM of the system. The remaining three coordinates were used to define the magnitude and orientation of the vector connecting the CMs of the fragments using the standard spherical polar format, the angles being measured with respect to the same space-fixed axes. The magnitude of the vector, i.e., the distance between the two fragments, was taken to be the reaction coordinate.

In paper I an alternative allocation was used which differs from that of Klippenstein and Marcus in the choice of angles to describe the relative orientation of the fragments. The internal modes of the fragments were identified as the conserved modes, and three coordinates were assigned to the overall CM (these coordinates are cyclic, factor out of all subsequent expressions, and will not be discussed further). A body-fixed set of axes was specified, such that the z axis was along the line connecting the fragment CMs. The orientation of this set of body-fixed axes relative to the space-fixed was defined using three Euler angles. This leaves six coordinates to be assigned. One of these was assigned to the distance between the fragment CMs and was used as the reaction coordinate. The remaining five degrees of freedom describe the orientation of the fragments *relative to the body-fixed frame*. The advantage of this approach is that in the absence of an external field the Euler angles describing the overall orientation of body-fixed frame are cyclic and can be integrated over analytically as will be shown below. Also, the potential describing the interaction of the fragments is most naturally given in terms of body-fixed coordinates, making it more amenable to numerical integration schemes. The disadvantage of a body-fixed approach is that Coriolis terms appear in the Hamiltonian. As will be shown later, however, such terms do not preclude an analytical evaluation of the

momentum integrals involved in the canonical or microcanonical rate expressions. The body-fixed approach will be used in this paper. The coordinates for the case of two associating nonlinear tops and will be described in more detail below.

The details of the allocation of degrees of freedom for the various combinations of fragment types are as follows. All fragments are modeled as one of the following types of rigid tops: atom, linear, or nonlinear. There are thus five pairwise combinations, excluding the atom+atom case. Let the total number of atoms in both fragments be N . There are three external rotation coordinates $\mathbf{q}_{\text{ext}} = (q_1, q_2, q_3)$ and m internal angles $\mathbf{q}_{\text{int}} = (q_4, q_5, \dots, q_n)$ describing the relative orientation of the two (rigid) fragments, where $n = m + 3$ is the number of transitional/external rotational coordinates. One coordinate, the reaction coordinate, specifies the separation of the fragments and is designated q_{n+1} . The $n+1$ -dimensional coordinate vector in terms of which the inter-fragment Hamiltonian H_{tr} will be expressed is $\mathbf{q} = (\mathbf{q}_{\text{ext}}, \mathbf{q}_{\text{int}}, q_{n+1}) = (q_1, q_2, \dots, q_{n+1})$. The number of conserved modes, $N_{\text{cons}} = 3N - 3 - (n+1)$, is equal to the number of vibrational degrees of freedom in the two fragments. A significant feature is that there are only five different cases to consider and, once case-specific formulas for a general reaction coordinate are obtained for each, the overall problem is solved in a formal sense. Case-specific results are compiled in the companion paper and emanate from the general formal expression derived herein.

C. General canonical rate constant expression

All derivations in Secs. III and IV will be done at the canonical (temperature-resolved) level. There are several reasons for this choice. It makes these sections compatible with our previous work on CFTST. The notation and mathematical development is somewhat simpler at the canonical level than at the microcanonical level. All formal development in the next two sections could have been done at the microcanonical level but we have instead deferred this treatment until Sec. V where we exploit Laplace transformation to obtain the microcanonical rate expression from its canonical counterpart.

In general the canonical rate coefficient as a function of the reaction coordinate, R , can be written as^{19,20}

$$k(T, R) = g_e g_\sigma^{-1} \frac{1}{Q_{\text{react}}(T)} \frac{1}{h^N} \times \int d\mathbf{q} d\mathbf{p} \delta(f(\mathbf{q}) - R) \dot{f}(\mathbf{q}) u(\dot{f}) e^{-\beta H}. \quad (1)$$

Q_{react} is the partition function of the reactants in the center-of-mass frame; g_e and g_σ are the ratios of transition state to reactant electronic degeneracy factors and symmetry factors, respectively. In the phase integral $\mathbf{q}$ refers to the generalized coordinates and $\mathbf{p}$ are the momenta conjugate to $\mathbf{q}$. The function $f(\mathbf{q})$ defines the DS separating reactants and products and u is the Heaviside unit step function. The overall Hamiltonian of the system is denoted H and $\beta = 1/k_B T$ where k_B is Boltzmann's constant.

The previous discussion outlined the distinction between conserved and transitional modes, the two groups being, to a good approximation, decoupled. This allows the overall Hamiltonian to be written as a sum, $H_c + H_{\text{tr}}$, H_c referring to the conserved modes. Also, in FTST, the DS is only a function of the transitional coordinates, i.e., $f(\mathbf{q}) = f(\mathbf{q}_{\text{tr}})$ since the assumed rigidity of each fragment precludes the DS from depending explicitly on the conserved mode (internal vibrational) coordinates. This allows the above phase integral to be factored into two integrals: one over the conserved coordinates and the other over transitional coordinates. The integral over the conserved coordinates will be denoted Q_{cons} and is the canonical partition function for these modes. Standard expressions are available for Q_{cons} and so it will not be discussed further. With this factorization the rate constant expression becomes,

$$k(T, R) = g_e g_\sigma^{-1} \frac{Q_{\text{cons}}(T)}{Q_{\text{react}}(T)} \frac{1}{h^{n+1}} \times \int d\mathbf{q}_{\text{tr}} d\mathbf{p}_{\text{tr}} \delta(R - f(\mathbf{q}_{\text{tr}})) \dot{f}(\mathbf{q}_{\text{tr}}) u(\dot{f}) e^{-\beta H_{\text{tr}}}, \quad (2)$$

where $n+1$ is the number of transitional/external rotational modes, including the RC. This expression can be written in a compact and insightful form by identifying the canonical transitional mode flux $\mathcal{F}$ through the DS orthogonal to the reaction coordinate,

$$\mathcal{F}(T, R) = \frac{1}{h^{n+1}} \int d\mathbf{q}_{\text{tr}} d\mathbf{p}_{\text{tr}} \delta(R - f(\mathbf{q}_{\text{tr}})) \dot{f}(\mathbf{q}_{\text{tr}}) u(\dot{f}) e^{-\beta H_{\text{tr}}} \equiv \frac{kT}{h} Q_{\text{tm}}(T, R), \quad (3)$$

where Q_{tm} is the transitional mode partition function for the n transitional/external rotational degrees of freedom. Equation (2) is thus rewritten

$$k(T, R) = g_e g_\sigma^{-1} \frac{Q_{\text{cons}}(T, R) \mathcal{F}(T, R)}{Q_{\text{react}}(T)} = g_e g_\sigma^{-1} \frac{kT}{h} \frac{Q_{\text{cons}}(T, R) Q_{\text{tm}}(T, R)}{Q_{\text{react}}(T)}. \quad (4)$$

The final rate coefficient is determined by optimizing Eq. (4) with respect to R giving

$$k(T) = g_e g_\sigma^{-1} \frac{kT}{h} \frac{Q_{\text{cons}}^\ddagger(T) Q_{\text{tm}}^\ddagger(T)}{Q_{\text{react}}(T)}. \quad (5)$$

The “ $\ddagger$ ” on Q_{cons} and Q_{tm} indicates that these quantities are evaluated at that value of R , called $R^\ddagger$, which minimizes their product at the given temperature, i.e., minimizes the numerator of the rate constant expression in Eq. (4). It is customary to rewrite Eq. (5) as

$$k(T) = g_e g_\sigma^{-1} \frac{kT}{h} \frac{Q_{\text{cons}}^\ddagger(T) Q_{\text{tm}}^\ddagger(T)}{Q_{\text{react}}(T)} e^{-\beta V^\ddagger}, \quad (6)$$

where $V^\ddagger$ denotes the minimum potential energy consistent with the fragments being separated by $R^\ddagger$ for a particular choice of RC definition. That is, the configuration associated with $V^\ddagger$ is a “point” on the MEP, where a point is the set of

TABLE I. Fragment combinations and angles.

Fragments	m	γ_1	γ_2	γ_3	γ_4	γ_5
atom+linear top	1	θ_1				
atom+nonlinear top	2	θ_1	ϕ_1			
linear top+linear top	3	θ_1	θ_2	χ		
linear top+nonlinear top	4	θ_1	ϕ_1	θ_2	χ	
nonlinear top+nonlinear top	5	θ_1	ϕ_1	θ_2	ϕ_2	χ

Cartesian coordinates $\mathbf{r}_{i,\text{MEP}}$ specifying the positions of all atoms. The particular MEP point is $\mathbf{r}'_{i,\text{MEP}}$ such that the RC value is $R^\dagger = f(\mathbf{r}'_{i,\text{MEP}})$. The factor $e^{-\beta V^\dagger}$ indicates that reactants and transition state have a common energy origin and that potential energy functions appearing in $Q_{\text{cons}}^\dagger$ and $Q_{tm}^\dagger$ are implicitly evaluated with respect to $V^\dagger$.

In Secs. III and IV, we shall focus exclusively on deriving expressions for $\mathcal{F}$.

III. CFTST FOR CENTERS-OF-MASS REACTION COORDINATE

For a CMRC, a convenient set of $n+1$ coordinates for transitional and external degrees of freedom plus the reaction coordinate is

$$\mathbf{q}_{\text{tr}} = \mathbf{q}_{\text{cm}} = (\mathbf{q}_{\text{cm,ext}}, \mathbf{q}_{\text{cm,int}}, \mathbf{q}_{\text{cm},n+1}). \quad (7)$$

Here $q_{\text{cm},n+1}$ is the separation of the CMs of the two fragments. $\mathbf{q}_{\text{cm,ext}} = (\Phi_{\text{cm}}, \Theta_{\text{cm}}, \Psi_{\text{cm}})$ are the Euler angles which orient a body-fixed xyz Cartesian coordinate system, whose origin is at the overall center of mass, with respect to a space-fixed XYZ system. There are a number of ways to specify how the xyz axes are fixed to the molecular system—in this paper we use the specification given in paper I. Briefly, the z-axis is aligned with the separation of the CMs ($q_{\text{cm},n+1}$) and the y-axis is aligned with the line of nodes of fragment 1. The internal coordinates $\mathbf{q}_{\text{cm,int}} = \boldsymbol{\gamma} = (\gamma_1, \dots, \gamma_m)$ are Euler angles specifying the relative orientation of fragments and are defined with respect to the body-fixed xyz axes. Table I shows the number and type of internal angles for each combination of fragment types; the subscripts on θ and ϕ refer to fragments 1 and 2. As in paper I, we adopt the Zare convention²¹ in which the orientation of the principal axes of a fragment with respect to the body-fixed xyz system is specified by three angles: counterclockwise rotations first by ϕ about the z-axis, then by θ about the y-axis, and finally by χ about the z-axis. Since the y-axis is the line of nodes of fragment 1, no final rotation about the z-axis is needed and χ , which is associated with fragment 2, is actually the relative polar angle (or dihedral angle) about $q_{\text{cm},n+1}$ (z-axis).

In the body-fixed frame the Hamiltonian for overall rotation plus transitional mode motion plus RC motion along $q_{\text{cm},n+1}$ (total of $n+1$ degrees of freedom) is

$$H_{\text{tr}} = T + V_{\text{tr}}(\mathbf{q}_{\text{cm}}), \quad (8)$$

where T is the kinetic energy which can be written²²

$$2T = \sum_{i=1}^N m_i \dot{\mathbf{r}}_i \cdot \dot{\mathbf{r}}_i + 2\omega \cdot \sum_{i=1}^N m_i \mathbf{r}_i \times \dot{\mathbf{r}}_i + \omega^T \cdot \mathbf{I} \cdot \omega. \quad (9)$$

Here m_i is the mass of atom i ; $\mathbf{r}_i = (x_i, y_i, z_i)$ is the position vector of atom i in the body-fixed frame; $\omega = (\omega_x, \omega_y, \omega_z)$ are the instantaneous angular velocities about the x, y, and z axes for rotation of the entire system; and $\mathbf{I}$ is the moment of inertia tensor for the entire system in the body-fixed xyz system. A consequence of the division of coordinates between conserved and transitional modes is that $\mathbf{r}_i$ can be written in terms of the transitional coordinates specified below and so implicit in Eq. (9) is the constraint that the two fragments are rigid.

To evaluate the flux integral in Eq. (5) the kinetic energy must be written in terms of generalized velocities, which, for the transitional modes are

$$\dot{\mathbf{q}}_{\text{cm}} = (\dot{\mathbf{q}}_{\text{cm,ext}}, \dot{\mathbf{q}}_{\text{cm,int}}, \dot{q}_{\text{cm},n+1}). \quad (10)$$

A related set of velocities that will be useful, but which are not generalized velocities, is

$$\dot{\mathbf{Q}}_{\text{cm}} = (\dot{\mathbf{Q}}_{\text{cm,ext}}, \dot{\mathbf{Q}}_{\text{cm,int}}, \dot{q}_{\text{cm},n+1}), \quad (11)$$

where $\dot{\mathbf{Q}}_{\text{cm,ext}} = (\omega_x, \omega_y, \omega_z)$. The kinetic energy can also be expressed in terms of $\dot{\mathbf{q}}_{\text{cm}}$ as

$$2T = \dot{\mathbf{q}}_{\text{cm}}^T \cdot \mathbf{a}^0 \cdot \dot{\mathbf{q}}_{\text{cm}}. \quad (12)$$

The superscript “0” on $\mathbf{a}^0$ indicates a CMRC; this label will be dropped in the next section where $\mathbf{a}$ will then denote the kinetic energy matrix for an arbitrary reaction coordinate.

In terms of $\dot{\mathbf{Q}}_{\text{cm}}$, we have

$$2T = \dot{\mathbf{Q}}_{\text{cm}}^T \cdot \mathbf{A}^0 \cdot \dot{\mathbf{Q}}_{\text{cm}}. \quad (13)$$

The quantity $\mathbf{A}^0$ is the kinetic energy matrix, widely known as the inverse G -matrix. The two expressions for the kinetic energy, Eqs. (12) and (13) are related because $\dot{\mathbf{Q}}_{\text{cm}} = \dot{\mathbf{S}} \mathbf{q}_{\text{cm}}$ and so the associated kinetic energy matrices are related by $\mathbf{a}^0 = \mathbf{S}^T \mathbf{A}^0 \mathbf{S}$. The matrix $\mathbf{S}$ is a similarity transformation,

$$\mathbf{S} = \begin{pmatrix} \mathbf{s}_{3 \times 3} & \mathbf{0}_{3 \times m} & \mathbf{0}_{3 \times 1} \\ \mathbf{0}_{m \times 3} & \mathbf{1}_{m \times m} & \mathbf{0}_{m \times 1} \\ \mathbf{0}_{1 \times 3} & \mathbf{0}_{1 \times m} & 1 \end{pmatrix}, \quad (14)$$

where $\mathbf{s}$ is the well known transformation^{22,23} connecting $(\omega_x, \omega_y, \omega_z)$ and $(\dot{\Phi}_{\text{cm}}, \dot{\Theta}_{\text{cm}}, \dot{\Psi}_{\text{cm}})$. In the y-axis representation of rotations it has the form:²²

$$\mathbf{s} = \begin{bmatrix} -\sin \Theta \cos \Psi & \sin \Psi & 0 \\ \sin \Theta \sin \Psi & \cos \Psi & 0 \\ \cos \Theta & 0 & 1 \end{bmatrix}. \quad (15)$$

The penultimate result of this section involves the determinant $|\mathbf{a}^0|$, whereas the final result is expressed in terms of $|\mathbf{A}^0|$. These two determinants are related via

$$|\mathbf{a}^0| = |\mathbf{S}^T \mathbf{A}^0 \mathbf{S}| = |\mathbf{S}|^2 |\mathbf{A}^0| = |\mathbf{s}|^2 |\mathbf{A}^0| = (\sin \Theta)^2 |\mathbf{A}^0|. \quad (16)$$

We note in passing that both $\mathbf{a}^0$ and $\mathbf{A}^0$ are symmetric.

The decision to use $\mathbf{A}^0$ is based on its simpler form: the upper left corner is the inertia tensor $\mathbf{I}$ whereas the corresponding corner of $\mathbf{a}^0$ contains linear combinations of the

elements of $\mathbf{I}$. The elements of $\mathbf{A}^0$ can be obtained by writing each x_i , y_i , and z_i in terms of $(\dot{\gamma}, q_{\text{cm},n+1})$, substituting in Eq. (9), and regrouping terms to obtain an expression in the form of Eq. (13). The precise form of the matrix elements A_{ij}^0 has no bearing on the general derivation presented herein; some of the simpler cases have been presented in papers I and II and a more detailed discussion is deferred to the companion paper (paper IV).

The general form of $\mathbf{A}^0$ is

$$\mathbf{A}^0 = \begin{pmatrix} \mathbf{I}_{3 \times 3} & (\mathbf{A}_{\text{int-ext}}^0)_{3 \times m} & \mathbf{0}_{3 \times 1} \\ (\mathbf{A}_{\text{int-ext}}^0)^T_{m \times 3} & (\mathbf{A}_{\text{int}}^0)_{m \times m} & \mathbf{0}_{m \times 1} \\ \mathbf{0}_{1 \times 3} & \mathbf{0}_{1 \times m} & \mu \end{pmatrix},$$

$$\text{where } \mathbf{I} = \begin{pmatrix} I_{xx} & I_{xy} & I_{xz} \\ I_{xy} & I_{yy} & I_{yz} \\ I_{xz} & I_{yz} & I_{zz} \end{pmatrix} \quad (17)$$

and

$$\mathbf{A}_{\text{int-ext}}^0 = \begin{pmatrix} A_{1x}^0 & A_{2x}^0 & \cdots & A_{mx}^0 \\ A_{1y}^0 & \cdots & & A_{my}^0 \\ A_{1z}^0 & \cdots & & A_{mz}^0 \end{pmatrix},$$

and

$$\mathbf{A}_{\text{int}}^0 = \begin{pmatrix} A_{11}^0 & A_{12}^0 & \cdots & A_{1m}^0 \\ A_{12}^0 & \ddots & & \vdots \\ \vdots & & & \\ A_{1m}^0 & \cdots & & A_{mm}^0 \end{pmatrix}. \quad (18)$$

A distinctive feature of the kinetic energy matrix for the CMRC is the absence of coupling between the RC, $q_{\text{cm},n+1} = R_{\text{cm}}$, and all other n degrees of freedom. Note that $\mathbf{I}$ and $\mathbf{A}_{\text{int}}^0$ are symmetric and that the form of $\mathbf{S}$ is such that $\mathbf{a}_{\text{int}}^0 = \mathbf{A}_{\text{int}}^0$ and $\mathbf{a}_{\text{int-ext}}^0 = \mathbf{A}_{\text{int-ext}}^0$.

In FTST with a CMRC, the DS is given by $f(\mathbf{q}_{\text{cm}}) = q_{\text{cm},n+1}$. Comparison of Eqs. (2) and (4) shows that the canonical flux at a particular value R_{cm} of $q_{\text{cm},n+1}$ is given by

$$\begin{aligned} \mathcal{F}^0(R_{\text{cm}}) &= h^{-(n+1)} \int \cdots \int d\mathbf{q}_{\text{cm}} \int \cdots \int d\mathbf{p}_{\text{cm}} \\ &\quad \times \delta(q_{\text{cm},n+1} - R_{\text{cm}}) \dot{q}_{\text{cm},n+1} u(\dot{q}_{\text{cm},n+1}) e^{-\beta H_{\text{tr}}} \\ &= h^{-(n+1)} \int \cdots \int d\mathbf{q}_{\text{cm}} \delta(q_{\text{cm},n+1} - R_{\text{cm}}) e^{-\beta V_{\text{tr}}} \\ &\quad \times \int \cdots \int d\mathbf{p}_{\text{cm}} \dot{q}_{\text{cm},n+1} u(\dot{q}_{\text{cm},n+1}) e^{-\beta T}, \quad (19) \\ &= h^{-(n+1)} \int \cdots \int d\mathbf{q}_{\text{cm}} \delta(q_{\text{cm},n+1} - R_{\text{cm}}) e^{-\beta V_{\text{tr}}} \mathcal{P}^0, \end{aligned}$$

where $\mathcal{P}^0$ is the multidimensional momentum integral. In the remainder of this section we evaluate $\mathcal{P}^0$ analytically and reduce the expression for $\mathcal{F}^0$ to a configuration integral over just the internal Jacobi angles γ . The analytic evaluation of $\mathcal{P}^0$ at the canonical level parallels the derivation of Marcus in 1964²⁴ and is analogous to Klippenstein's 1994 derivation at

the microcanonical (E -resolved) level.¹⁹ We provide the complete derivation here not because it is completely new but because it introduces notation and concepts to be used in Sec. IV and because it provides several details not included in our earlier treatments.

In order to evaluate

$$\mathcal{P}^0 = \int d\mathbf{p}_{\text{cm}} \dot{q}_{\text{cm},n+1} u(\dot{q}_{\text{cm},n+1}) e^{-\beta T} \quad (20)$$

one must express the integrand in terms of $\mathbf{p}_{\text{cm}}$. The kinetic energy expression is

$$2T = \dot{\mathbf{q}}_{\text{cm}}^T \cdot \mathbf{a}^0 \cdot \dot{\mathbf{q}}_{\text{cm}} = \sum_{i,j=1}^{n+1} a_{ij}^0 \dot{q}_{\text{cm},i} \dot{q}_{\text{cm},j}. \quad (21)$$

Let $L = T - V$ be the Lagrangian for transitional mode motion. Then

$$p_{\text{cm},i} = \frac{\partial L}{\partial \dot{q}_{\text{cm},i}} = \frac{\partial T}{\partial \dot{q}_{\text{cm},i}} = \sum_{j=1}^{n+1} a_{ij}^0 \dot{q}_{\text{cm},j} \quad \text{for } i=1, \dots, n+1. \quad (22)$$

In matrix notation, we have

$$\mathbf{p}_{\text{cm}} = \mathbf{a}^0 \dot{\mathbf{q}}_{\text{cm}} \quad \text{or} \quad \dot{\mathbf{q}}_{\text{cm}} = (\mathbf{a}^0)^{-1} \mathbf{p}_{\text{cm}} \equiv \mathbf{g}^0 \mathbf{p}_{\text{cm}}. \quad (23)$$

Thus,

$$2T = (\mathbf{g}^0 \mathbf{p}_{\text{cm}})^T \mathbf{a}^0 (\mathbf{g}^0 \mathbf{p}_{\text{cm}}) = \mathbf{p}_{\text{cm}}^T (\mathbf{g}^0)^T \mathbf{a}^0 \mathbf{g}^0 \mathbf{p}_{\text{cm}} = \mathbf{p}_{\text{cm}}^T \mathbf{g}^0 \mathbf{p}_{\text{cm}}, \quad (24)$$

since $\mathbf{a}^0 \mathbf{g}^0 = \mathbf{1}$ and $(\mathbf{g}^0)^T = \mathbf{g}^0$. The integral over $p_{\text{cm},n+1}$ in $\mathcal{P}^0$ is evaluated first by expressing $\dot{q}_{\text{cm},n+1}$ and $u(\dot{q}_{\text{cm},n+1})$ in the integrand of Eq. (20) in terms of $p_{\text{cm},n+1}$:

$$\dot{q}_{\text{cm},n+1} = \frac{\partial H_{\text{tr}}}{\partial p_{\text{cm},n+1}} = \frac{\partial T}{\partial p_{\text{cm},n+1}} = \sum_{j=1}^{n+1} g_{n+1,j}^0 p_{\text{cm},j}. \quad (25)$$

The Heaviside function in Eq. (20) is nonzero when $\dot{q}_{\text{cm},n+1} > 0$ and this is satisfied when,

$$p_{\text{cm},n+1} > p_{\text{cm},n+1}^*, \quad (26)$$

where

$$p_{\text{cm},n+1}^* = \frac{-\sum_{j=1}^n g_{n+1,j}^0 p_{\text{cm},j}}{g_{n+1,n+1}^0} \quad (27)$$

is the solution of Eq. (22) for $\dot{q}_{\text{cm},n+1} = 0$. The momentum integral becomes

$$\begin{aligned} \mathcal{P}^0 &= \int \cdots \int d\mathbf{p}_{\text{cm},r} \int_{p_{n+1}^*}^{\infty} dp_{\text{cm},n+1} \frac{\partial T}{\partial p_{\text{cm},n+1}} e^{-\beta T} \\ &= \int \cdots \int d\mathbf{p}_{\text{cm},r} \int_{T^{0*}}^{\infty} dT e^{-\beta T} \\ &= \beta^{-1} \int \cdots \int d\mathbf{p}_{\text{cm},r} e^{-\beta T^{0*}}, \quad (28) \end{aligned}$$

where $d\mathbf{p}_{\text{cm},r} = dp_{\text{cm},1} \cdots dp_{\text{cm},n}$ is the volume element associated with the *reduced* momentum and T^{0*} is a constrained kinetic energy:

$$\begin{aligned} T^{0*} &= T(\dot{q}_{\text{cm},1}, \dots, \dot{q}_{\text{cm},n}; \dot{q}_{\text{cm},n+1} = 0) \\ &= T(\dot{p}_{\text{cm},1}, \dots, \dot{p}_{\text{cm},n}; p_{\text{cm},n+1}^*). \quad (29) \end{aligned}$$

Substituting Eq. (27) for $p_{\text{cm},n+1}^*$ into the kinetic energy expression [Eq. (24)] yields

$$2T^{0*} = \sum_{i,j=1}^n \left[g_{ij}^0 - \frac{g_{i,n+1}^0 g_{n+1,j}^0}{g_{n+1,n+1}^0} \right] p_{\text{cm},i} p_{\text{cm},j} = \mathbf{p}_{\text{cm},r}^T \mathbf{g}_r^0 \mathbf{p}_{\text{cm},r}, \quad (30)$$

where $\mathbf{g}_r^0$ is a $n \times n$ matrix which is reduced in the sense that its dimensions are one less than those of $\mathbf{g}^0$ and whose elements are

$$(g_r^0)_{ij} = g_{ij}^0 - \frac{g_{i,n+1}^0 g_{n+1,j}^0}{g_{n+1,n+1}^0}, \quad (i,j) = 1, \dots, n. \quad (31)$$

In terms of $\mathbf{p}_{\text{cm},r}$ and $\mathbf{g}_r^0$ the momentum integral is

$$\mathcal{P}^0 = \beta^{-1} \int_{-\infty}^{+\infty} \dots \int_{-\infty}^{+\infty} d\mathbf{p}_{\text{cm},r} e^{-(\beta/2)(\mathbf{p}_{\text{cm},r}^T \mathbf{g}_r^0 \mathbf{p}_{\text{cm},r})} \quad (32)$$

and the integrals can be done analytically¹⁹ yielding

$$\mathcal{P}^0 = \beta^{-1} \left(\frac{2\pi}{\beta} \right)^{n/2} |\mathbf{g}_r^0|^{-1/2}. \quad (33)$$

In the special case of a CMRC, the kinetic energy associated with this coordinate is separable, i.e., $g_{i,n+1}^0 = g_{n+1,i}^0 = 0$ for $i = 1, \dots, n$ and $\mathbf{g}_r^0$ is obtained from $\mathbf{g}^0$ by removing the $(n+1)$ st row and $(n+1)$ st column from the latter. We note in advance that the corresponding full and reduced matrices for an arbitrary RC are not so simply related.

Since the kinetic energy is more readily, and perhaps more intuitively, expressed in terms of generalized velocities than conjugate momenta, we seek to express $\mathcal{P}^0$ in terms of the matrix $\mathbf{a}_r^0$ associated with the reduced velocity vector $\dot{\mathbf{q}}_{\text{cm},r}$. In terms of the corresponding generalized velocities the constrained kinetic energy [Eq. (30)] is

$$2T^{0*} = \sum_{i,j=1}^n a_{ij}^0 \dot{q}_{\text{cm},i} \dot{q}_{\text{cm},j} = \dot{\mathbf{q}}_{\text{cm},r}^T \mathbf{a}_r^0 \dot{\mathbf{q}}_{\text{cm},r}, \quad (34)$$

where $\dot{\mathbf{q}}_{\text{cm},r} = (\dot{q}_{\text{cm},1}, \dots, \dot{q}_{\text{cm},n})$ and $\mathbf{a}_r^0$ is a $n \times n$ reduced matrix obtained by striking the $(n+1)$ st row and $(n+1)$ st column of $\mathbf{a}^0$. For the CMRC, the relation between $\mathbf{a}_r^0$ and $\mathbf{g}_r^0$ is simple, since $\mathbf{g}^0$ is the inverse of $\mathbf{a}^0$ and $\mathbf{a}^0$ is block diagonal,

$$\mathbf{a}^0 = \begin{bmatrix} (\mathbf{a}_r^0)_{n \times n} & \mathbf{0}_{n \times 1} \\ \mathbf{0}_{1 \times n} & \mu \end{bmatrix}, \quad (35)$$

then $\mathbf{g}^0$ will have the same block structure and each block will be the inverse of the corresponding block of $\mathbf{a}^0$:

$$\mathbf{g}^0 = \begin{bmatrix} (\mathbf{g}_r^0)_{n \times n} = (\mathbf{a}_r^0)^{-1}_{n \times n} & \mathbf{0}_{n \times 1} \\ \mathbf{0}_{1 \times n} & \mu^{-1} \end{bmatrix}. \quad (36)$$

Since $|\mathbf{g}_r^0 \mathbf{a}_r^0| = |\mathbf{g}_r^0| |\mathbf{a}_r^0| = 1$,

$$\mathcal{P}^0 = \beta^{-1} \left(\frac{2\pi}{\beta} \right)^{n/2} |\mathbf{a}_r^0|^{1/2}. \quad (37)$$

The form of the matrix $\mathbf{S}$ relating $\mathbf{a}^0$ and $\mathbf{A}^0$ [Eq. (14)] ensures that the determinants of the corresponding reduced matrices are related in exactly the same way as full matrix determinants [Eq. (16)]. Thus

$$\mathcal{P}^0 = \beta^{-1} \left(\frac{2\pi}{\beta} \right)^{n/2} \sin \Theta |\mathbf{A}_r^0|^{1/2}. \quad (38)$$

The flux integral [Eq. (19)] becomes

$$\mathcal{F}^0(R_{\text{cm}}) = \frac{kT}{h} \left(\frac{2\pi}{\beta h^2} \right)^{n/2} \int \dots \int d\mathbf{q}_{\text{cm}} \times \delta(q_{\text{cm},n+1} - R_{\text{cm}}) \sin \Theta |\mathbf{A}_r^0|^{1/2} e^{-\beta V_{\text{tr}}}. \quad (39)$$

The first three components of $\mathbf{q}_{\text{cm}}$ are the “external” angles Φ , Θ , and Ψ . In the absence of an external field the potential V_{tr} does not depend on the orientation of the entire system relative to the lab frame. In the companion paper, our analytic results for $|\mathbf{A}_r^0|$ reveal that it also does not depend on the external angle coordinates. Thus the integrals over these coordinates can be done analytically:

$$\begin{aligned} \mathcal{F}^0(R_{\text{cm}}) &= \frac{kT}{h} \left(\frac{2\pi}{\beta h^2} \right)^{n/2} \int_0^{2\pi} d\Phi \int_0^\pi d\Theta \sin \Theta \\ &\times \int_0^{2\pi} d\Psi \int \dots \int d\mathbf{q}_{\text{cm,int}} dq_{\text{cm},n+1} \\ &\times \delta(q_{\text{cm},n+1} - R_{\text{cm}}) |\mathbf{A}_r^0|^{1/2} e^{-\beta V_{\text{tr}}(\mathbf{q}_{\text{cm,int}}, q_{\text{cm},n+1})} \\ &= \frac{kT}{h} 8\pi^2 \left(\frac{2\pi}{\beta h^2} \right)^{n/2} \int \dots \int d\mathbf{q}_{\text{cm,int}} \\ &\times |\mathbf{A}_r^0|^{1/2} e^{-\beta V_{\text{tr}}(\mathbf{q}_{\text{cm,int}}, R_{\text{cm}})}. \end{aligned} \quad (40)$$

Thus the flux integral has been reduced to a m -dimensional configuration integral, where $m = 1, \dots, 5$ is the number of internal coordinates (see Table I). Equation (40) is identical to the result obtained in paper I. Note that $|\mathbf{A}_r^0|$ is a function of $\mathbf{q}_{\text{cm,int}}$ and R_{cm} , the precise form depending on the value of m . The explicit expressions for $m = 5$ and $m = 1$ are given in papers I and II, respectively. Explicit expressions for all values of m are provided in a companion paper.

IV. CFTST FOR VARIABLE REACTION COORDINATE

As discussed above, for a variable reaction coordinate the DS is defined as $f = f(\mathbf{q}_{\text{cm}})$ and so the canonical flux expression for a variable reaction coordinate (VRC) corresponding to that for a CMRC [Eq. (19)] is

$$\mathcal{F}(R) = h^{-(n+1)} \int \dots \int d\mathbf{q}_{\text{cm}} \int \dots \int d\mathbf{p}_{\text{cm}} \times \delta(f(\mathbf{q}_{\text{cm}}) - R) \dot{f} u(\dot{f}) e^{-\beta H_{\text{tr}}} \quad (41)$$

for a particular value R of the reaction coordinate distance. To facilitate the evaluation of Eq. (41) while retaining the conceptual and computational advantages of using external and internal angles from the centers-of-mass treatment, a new set of $n+1$ coordinates is introduced:

$$\mathbf{q} = (\mathbf{q}_{\text{cm,ext}}, \mathbf{q}_{\text{cm,int}}, q_{n+1}). \quad (42)$$

Here q_{n+1} is the separation of arbitrary fixed points in each fragment. Each fixed point is in general displaced from the fragment CM; particular displacements (one in each fragment) constitutes a particular choice of reaction coordinate.

Note that previous internal and external angle definitions are retained but the CM displacement is replaced by a displacement along the VRC.

Let $\mathbf{p}$ be the momentum vector conjugate to $\mathbf{q}$. To evaluate this flux integral, we need to express the kinetic energy in terms of $\mathbf{p}$ and change the variables of integration from $(\mathbf{q}_{\text{cm}}, \mathbf{p}_{\text{cm}})$ to $(\mathbf{q}, \mathbf{p})$.

Since q_{n+1} is the reaction coordinate we can identify it with the DS:

$$q_{n+1} = f(\mathbf{q}_{\text{cm, int}}; q_{\text{cm}, n+1}), \quad (43)$$

which can be inverted to give

$$q_{\text{cm}, n+1} = F(\mathbf{q}_{\text{cm, int}}; q_{n+1}). \quad (44)$$

In what follows we shall employ the transformation between $\mathbf{q}_{\text{cm}}$ and $\mathbf{q}$. Let the Jacobian for this transformation, $\mathbf{q}_{\text{cm}} \rightarrow \mathbf{q}$, be $|\mathcal{J}_q|$, such that,

$$d\mathbf{q}_{\text{cm}} = |\mathcal{J}_q| d\mathbf{q}. \quad (45)$$

Since these coordinate systems differ only in their $(n+1)$ th components, the form of the transform matrix $\mathcal{J}_q$ is

$$\mathcal{J}_q = \begin{pmatrix} \mathbf{1}_{3 \times 3} & \mathbf{0}_{3 \times m} & \mathbf{0}_{3 \times 1} \\ \mathbf{0}_{m \times 3} & \mathbf{1}_{m \times m} & \mathbf{0}_{m \times 1} \\ \mathbf{0}_{1 \times 3} & \mathbf{b}_{1 \times m} & \frac{\partial F}{\partial q_{n+1}} \end{pmatrix}, \quad (46)$$

where

$$\mathbf{b}_{1 \times m} = \left(\frac{\partial F}{\partial q_{\text{cm, int}, 1}}, \dots, \frac{\partial F}{\partial q_{\text{cm, int}, m}} \right) \equiv \nabla_{\mathbf{q}_{\text{cm, int}}} F. \quad (47)$$

The determinant is

$$|\mathcal{J}_q| = \frac{\partial F}{\partial q_{n+1}} \equiv \alpha. \quad (48)$$

The transformation between $\mathbf{p}_{\text{cm}}$ and $\mathbf{p}$ is obtained by considering the kinetic energy expression and exploiting the known relation between the corresponding generalized velocities²⁴

$$\dot{\mathbf{q}}_{\text{cm}} = \mathcal{J}_q \dot{\mathbf{q}}. \quad (49)$$

Substituting this expression for $\dot{\mathbf{q}}_{\text{cm}}$ in Eq. (12) yields

$$2T = \dot{\mathbf{q}}^T \mathcal{J}_q^T \mathbf{a}^0 \mathcal{J}_q \dot{\mathbf{q}} = \dot{\mathbf{q}}^T \mathbf{a} \dot{\mathbf{q}}, \quad (50)$$

where $\mathbf{a} \equiv \mathcal{J}_q^T \mathbf{a}^0 \mathcal{J}_q$. T can be expressed in terms of $\mathbf{p}$ by following the same procedure used to express T in terms of $\mathbf{p}_{\text{cm}}$ in Eqs. (21)–(24), i.e., by forming the Lagrangian

$$\mathbf{p} = \mathbf{a} \dot{\mathbf{q}} \quad \text{or} \quad \dot{\mathbf{q}} = \mathbf{a}^{-1} \mathbf{p} \equiv \mathbf{g} \mathbf{p}, \quad (51)$$

and

$$2T = (\mathbf{g} \mathbf{p})^T \mathbf{a} (\mathbf{g} \mathbf{p}) = \mathbf{p}^T \mathbf{g}^T \mathbf{a} \mathbf{g} \mathbf{p} = \mathbf{p}^T \mathbf{g} \mathbf{p}, \quad (52)$$

since $\mathbf{a} \mathbf{g} = \mathbf{1}$ and $\mathbf{g}^T = \mathbf{g}$. Equations (51) and (49) imply that

$$\mathbf{p} = \mathcal{J}_q^T \mathbf{a}^0 \mathcal{J}_q \dot{\mathbf{q}} = \mathcal{J}_q^T \mathbf{a}^0 \dot{\mathbf{q}}_{\text{cm}} = \mathcal{J}_q^T \mathbf{p}_{\text{cm}}, \quad (53)$$

and therefore that

$$\mathbf{p}_{\text{cm}} = (\mathcal{J}_q^T)^{-1} \mathbf{p}. \quad (54)$$

The transformation from $(\mathbf{q}_{\text{cm}}, \mathbf{p}_{\text{cm}})$ to $(\mathbf{q}, \mathbf{p})$ can thus be written

$$\begin{pmatrix} \mathbf{q}_{\text{cm}} \\ \mathbf{p}_{\text{cm}} \end{pmatrix} = \mathcal{J} \begin{pmatrix} \mathbf{q} \\ \mathbf{p} \end{pmatrix} = \begin{pmatrix} \mathcal{J}_q & \mathbf{0} \\ \mathbf{0} & (\mathcal{J}_q^T)^{-1} \end{pmatrix} \begin{pmatrix} \mathbf{q} \\ \mathbf{p} \end{pmatrix}. \quad (55)$$

Since $|\mathcal{J}| = |\mathcal{J}_q| |\mathcal{J}_q^T|^{-1} = 1$, this is a canonical transformation, meaning that a new Hamiltonian is obtained with the kinetic energy defined by Eq. (52) and $d\mathbf{q}_{\text{cm}} d\mathbf{p}_{\text{cm}} = d\mathbf{q} d\mathbf{p}$. The flux integral [Eq. (41)] becomes

$$\begin{aligned} \mathcal{F}(R) &= h^{-(n+1)} \int \cdots \int d\mathbf{q} \int \cdots \int d\mathbf{p} \\ &\quad \times \delta(q_{n+1} - R) \dot{q}_{n+1} u(\dot{q}_{n+1}) e^{-\beta H_{\text{tr}}} \\ &= h^{-(n+1)} \int \cdots \int d\mathbf{q} \delta(q_{n+1} - R) e^{-\beta V_{\text{tr}}} \mathcal{P}, \end{aligned} \quad (56)$$

where the momentum integral is

$$\mathcal{P} = \int \cdots \int d\mathbf{p} \quad \dot{q}_{n+1} u(\dot{q}_{n+1}) e^{-\beta T}. \quad (57)$$

The evaluation of $\mathcal{P}$ exactly parallels that for $\mathcal{P}^0$ with $\dot{q}_{\text{cm}, n+1}$, $p_{\text{cm}, n+1}^*$, T^{0*} , and $\mathbf{g}_r^0$ in Eqs. (25)–(31) replaced, respectively, by $\dot{q}_{n+1}$, p_{n+1}^* , T^* , and $\mathbf{g}_r$, where

$$\dot{q}_{n+1} = \frac{\partial T}{\partial p_{n+1}} = \sum_{j=1}^{n+1} g_{n+1, j} p_j. \quad (58)$$

$$p_{n+1}^* = \frac{-\sum_{j=1}^n g_{n+1, j} p_j}{g_{n+1, n+1}} \quad (59)$$

$$2T^* = \sum_{i, j=1}^n \left[g_{ij} - \frac{g_{i, n+1} g_{n+1, j}}{g_{n+1, n+1}} \right] p_i p_j = \mathbf{p}_r^T \mathbf{g}_r^* \mathbf{p}_r, \quad (60)$$

and $\mathbf{g}_r$ is the $n \times n$ reduced version of $\mathbf{g}$ whose elements are

$$(g_r)_{ij} = g_{ij} - \frac{g_{i, n+1} g_{n+1, j}}{g_{n+1, n+1}}, \quad (i, j) = 1, \dots, n. \quad (61)$$

In Appendix A it is shown that $\mathbf{g}_r$ is the inverse of $\mathbf{a}_r$, where $\mathbf{a}_r$ is a $n \times n$ reduced matrix obtained by striking the $(n+1)$ st row and $(n+1)$ st column of $\mathbf{a}$. In general, $g_{i, n+1} = g_{n+1, i} \neq 0$ for $i = 1, \dots, n$ so that $\mathbf{g}_r$ is not obtained from $\mathbf{g}$ by simply removing the $(n+1)$ st row and $(n+1)$ st column from the latter [see Eq. (61)]. In terms of the corresponding generalized velocities the constrained kinetic energy (i.e., with $\dot{q}_{n+1} = 0$) is

$$2T^* = \sum_{i, j=1}^n a_{ij} \dot{q}_i \dot{q}_j = \dot{\mathbf{q}}_r^T \mathbf{a}_r \dot{\mathbf{q}}_r, \quad (62)$$

where $\dot{\mathbf{q}}_r = (\dot{q}_{\text{cm}, 1}, \dots, \dot{q}_{\text{cm}, n}) = \dot{\mathbf{q}}_{\text{cm}, r}$.

In terms of $\mathbf{p}_r$ and $\mathbf{g}_r$ the momentum integral is

$$\begin{aligned} \mathcal{P} &= \beta^{-1} \int_{-\infty}^{+\infty} \cdots \int_{-\infty}^{+\infty} d\mathbf{p}_r e^{-(\beta/2)(\mathbf{p}_r^T \mathbf{g}_r \mathbf{p}_r)} \\ &= \beta^{-1} \left(\frac{2\pi}{\beta} \right)^{n/2} |\mathbf{g}_r|^{-1/2}. \end{aligned} \quad (63)$$

Since $\mathbf{a}_r$ is the inverse of $\mathbf{g}_r$, then $|\mathbf{g}_r| = |\mathbf{a}_r|^{-1}$ and

$$\mathcal{P} = \beta^{-1} \left(\frac{2\pi}{\beta} \right)^{n/2} |\mathbf{a}_r|^{1/2}. \quad (64)$$

Before using the above expression for $\mathcal{P}$ in the evaluation of the configuration integral [Eq. (56)], we express $\mathbf{a}_r$ in terms of $\mathbf{A}_r^0$ and the elements of $\mathcal{J}_q$. One begins by expressing $\mathbf{a}$ in terms of $\mathbf{a}^0$. From Eq. (50)

$$\begin{aligned} \mathbf{a} &= \mathcal{J}_q^T \mathbf{a}^0 \mathcal{J}_q = \begin{pmatrix} \mathbf{1}_{3 \times 3} & \mathbf{0}_{3 \times m} & \mathbf{0}_{3 \times 1} \\ \mathbf{0}_{m \times 3} & \mathbf{1}_{m \times m} & \mathbf{b}_{m \times 1}^T \\ \mathbf{0}_{1 \times 3} & \mathbf{0}_{1 \times m} & \alpha \end{pmatrix} \begin{pmatrix} (\mathbf{a}_{\text{ext}}^0)_{3 \times 3} & (\mathbf{a}_{\text{int-ext}}^0)_{3 \times m} & \mathbf{0}_{3 \times 1} \\ (\mathbf{a}_{\text{int-ext}}^0)^T_{m \times 3} & (\mathbf{a}_{\text{int}}^0)_{m \times m} & \mathbf{0}_{m \times 1} \\ \mathbf{0}_{1 \times 3} & \mathbf{0}_{1 \times m} & \mu \end{pmatrix} \begin{pmatrix} \mathbf{1}_{3 \times 3} & \mathbf{0}_{3 \times m} & \mathbf{0}_{3 \times 1} \\ \mathbf{0}_{m \times 3} & \mathbf{1}_{m \times m} & \mathbf{0}_{m \times 1} \\ \mathbf{0}_{1 \times 3} & \mathbf{b}_{1 \times m} & \alpha \end{pmatrix} \\ &= \begin{pmatrix} (\mathbf{a}_{\text{ext}}^0)_{3 \times 3} & (\mathbf{a}_{\text{int-ext}}^0)_{3 \times m} & \mathbf{0}_{3 \times 1} \\ (\mathbf{a}_{\text{int-ext}}^0)^T_{m \times 3} & (\mathbf{a}_{\text{int}}^0)_{m \times m} + \mu(\mathbf{b}\mathbf{b}^T)_{m \times m} & \mu\alpha(\mathbf{b}^T)_{m \times 1} \\ \mathbf{0}_{1 \times 3} & \mu\alpha\mathbf{b}_{1 \times m} & \mu\alpha^2 \end{pmatrix}. \end{aligned} \quad (65)$$

Because the external Euler angles are the same in the mixed coordinates $\mathbf{q}$ and the original centers-of-mass coordinates $\mathbf{q}_{\text{cm}}$ [Eq. (42)], the matrices $\mathbf{a}_r$ and $\mathbf{A}_r$ are related in the same way as $\mathbf{a}_r^0$ and $\mathbf{A}_r^0$, i.e.,

$$\mathbf{a} = \mathbf{S}^T \mathbf{A} \mathbf{S}, \quad (66)$$

where $\mathbf{S}$ is given by Eq. (14). The form of $\mathbf{S}$ and the equivalences $\mathbf{a}_{\text{int}}^0 = \mathbf{A}_{\text{int}}^0$ and $\mathbf{a}_{\text{int-ext}}^0 = \mathbf{A}_{\text{int-ext}}^0$ noted in Sec. III yield

$$\mathbf{A} = \begin{pmatrix} \mathbf{I}_{3 \times 3} & (\mathbf{A}_{\text{int-ext}}^0)_{3 \times m} & \mathbf{0}_{3 \times 1} \\ (\mathbf{A}_{\text{int-ext}}^0)^T_{m \times 3} & (\mathbf{A}_{\text{int}}^0)_{m \times m} + \mu(\mathbf{b}\mathbf{b}^T)_{m \times m} & \mu\alpha(\mathbf{b}^T)_{m \times 1} \\ \mathbf{0}_{1 \times 3} & \mu\alpha\mathbf{b}_{1 \times m} & \mu\alpha^2 \end{pmatrix}. \quad (67)$$

The reduced version of $\mathbf{A}$ corresponding to $\mathbf{a}_r$ is obtained by striking the $(n+1)$ st row and $(n+1)$ st column of the former:

$$\begin{aligned} \mathbf{A}_r &= \begin{pmatrix} \mathbf{I}_{3 \times 3} & (\mathbf{A}_{\text{int-ext}}^0)_{3 \times m} \\ (\mathbf{A}_{\text{int-ext}}^0)^T_{m \times 3} & (\mathbf{A}_{\text{int}}^0)_{m \times m} + \mu(\mathbf{b}\mathbf{b}^T)_{m \times m} \end{pmatrix} \\ &= \begin{pmatrix} \mathbf{I}_{3 \times 3} & (\mathbf{A}_{\text{int-ext}}^0)_{3 \times m} \\ (\mathbf{A}_{\text{int-ext}}^0)^T_{m \times 3} & (\mathbf{A}_{\text{int}}^0)_{m \times m} \end{pmatrix} \\ &\quad + \begin{pmatrix} \mathbf{0}_{3 \times 3} & \mathbf{0}_{3 \times m} \\ \mathbf{0}_{m \times 3}^T & \mu(\mathbf{b}\mathbf{b}^T)_{m \times m} \end{pmatrix} \\ &\equiv \mathbf{A}_r^0 + \delta\mathbf{A}_r. \end{aligned} \quad (68)$$

In Appendix B, it is shown that $|\mathbf{A}_r|$ can be expressed in terms of the full and minor determinants of $\mathbf{A}_r^0$:

$$\begin{aligned} |\mathbf{A}_r^0 + \delta\mathbf{A}_r| &= |\mathbf{A}_r^0| + \sum_{i=4}^n (\delta\mathbf{A}_r)_{ii} (\mathbf{A}_r^0)^{ii} \\ &\quad + 2 \sum_{i=4}^{n-1} \sum_{j=i+1}^n (-1)^{i+j} (\delta\mathbf{A}_r)_{ij} (\mathbf{A}_r^0)^{ij}, \end{aligned} \quad (69)$$

where $(\mathbf{A}_r^0)^{ij}$ is the ij minor of $\mathbf{A}_r^0$. The form of the matrix $\mathbf{S}$ relating $\mathbf{a}$ to $\mathbf{A}$ means that the same relation exists between the determinants of the corresponding reduced matrices:

$$|\mathbf{a}_r| = (\sin \Theta)^2 |\mathbf{A}_r|. \quad (70)$$

The flux integral [Eq. (56)] becomes

$$\begin{aligned} \mathcal{F}(R) &= \frac{kT}{h} \left(\frac{2\pi}{\beta h^2} \right)^{n/2} \int \cdots \int d\mathbf{q}_{\text{cm,ext}} d\mathbf{q}_{\text{cm,int}} dq_{n+1} \\ &\quad \times \delta(q_{n+1} - R) \sin \Theta |\mathbf{A}_r|^{1/2} e^{-\beta V_{\text{tr}}} \\ &= \frac{kT}{h} 8\pi^2 \left(\frac{2\pi}{\beta h^2} \right)^{n/2} \int \cdots \int d\mathbf{q}_{\text{cm,int}} \\ &\quad \times |\mathbf{A}_r^0 + \delta\mathbf{A}_r|^{1/2} e^{-\beta V_{\text{tr}}(\mathbf{q}_{\text{cm,int}}, R)}. \end{aligned} \quad (71)$$

The flux integral for a VRC coordinate has thus been reduced to a m -dimensional configuration integral and is analogous to Eq. (40) for the CMRC case. Equation (71) is identical to the result obtained in paper II which started part way through the full derivation herein. We note that $|\mathbf{A}_r^0 + \delta\mathbf{A}_r|$ is a function of $\mathbf{q}_{\text{cm,int}}$ and R ; the detailed evaluation of this determinant for all five cases is the subject of a companion paper (paper IV).

V. MICROCANONICAL FTST FOR VARIABLE REACTION COORDINATE

The microcanonical FTST rate constant expression is

$$k(E) = g_e g_\sigma^{-1} \frac{N^\dagger(E)}{h \rho_{\text{react}}(E)}, \quad (72)$$

where g_σ and g_e are the same as in the corresponding canonical expression [Eq. (4)], $N^\dagger(E)$ is the transition state sum of states (number of states on the microcanonical DS with energy less than or equal to E), and $\rho_{\text{react}}(E)$ is the density of states of the reactants. The “ $\dagger$ ” on N indicates that the latter is evaluated at $R^\dagger$, that value of the reaction coordinate which minimizes the sum of states. In terms of

the conserved and transitional modes identified in FTST, the numerator of Eq. (72) can be written as the convolution

$$\begin{aligned} h^{-1}N^+(E') &= \int_0^{E'} d\epsilon \rho_{\text{cons}}(\epsilon, R^+) h^{-1}N_{\text{tr}}(E' - \epsilon, R^+) \\ &= \int_0^{E'} d\epsilon \rho_{\text{cons}}(\epsilon, R^+) \mathbf{f}(E' - \epsilon, R^+). \end{aligned} \quad (73)$$

Here $\rho_{\text{cons}}(\epsilon, R)$ is the conserved mode density of states at energy ϵ ; $E' = E - V^\ddagger$ is the total available energy relative to $V^\ddagger$ for the given definition and value of the RC; and $\mathbf{f}(\epsilon_{\text{tr}}, R)$ is the microcanonical transitional mode flux at energy $\epsilon_{\text{tr}} = E' - \epsilon$ through a DS orthogonal to the reaction coordinate. Note that both quantities depend on the value R of the reaction coordinate. The explicit expression for $\mathbf{f}$ for a variable reaction coordinate is obtained by replacing the Boltzmann factor $e^{-\beta H_{\text{tr}}}$ in the canonical flux expression [Eq. (56)] with $\delta(\epsilon_{\text{tr}} - H_{\text{tr}})$:

$$\begin{aligned} \mathbf{f}(\epsilon_{\text{tr}}, R) &= h^{-(n+1)} \int \cdots \int d\mathbf{q} \delta(q_{n+1} - R) \\ &\times \int \cdots \int d\mathbf{p} \dot{q}_{n+1} u(\dot{q}_{n+1}) \delta(\epsilon_{\text{tr}} - H_{\text{tr}}). \end{aligned} \quad (74)$$

$$\begin{aligned} \mathbf{f}(\epsilon_{\text{tr}}) &= \mathbf{f}(E' - \epsilon) = \mathcal{L}^{-1}\{\mathcal{F}(\beta)\} = \frac{8\pi^2}{h} \left(\frac{2\pi}{h^2}\right)^{n/2} \int \cdots \int d\mathbf{q}_{\text{cm, int}} |\mathbf{A}_r^0 + \delta\mathbf{A}_r|^{1/2} \mathcal{L}^{-1}\left\{\left(\frac{1}{\beta}\right)^{n/2+1} e^{-\beta V_{\text{tr}}}\right\} \\ &= \frac{8\pi^2}{h\Gamma\left(\frac{n}{2} + 1\right)} \left(\frac{2\pi}{h^2}\right)^{n/2} \int \cdots \int d\mathbf{q}_{\text{cm, int}} |\mathbf{A}_r^0 + \delta\mathbf{A}_r|^{1/2} (E' - \epsilon - V_{\text{tr}})^{n/2} u(E' - \epsilon - V_{\text{tr}}). \end{aligned} \quad (76)$$

The inverse transform in the third equality is found in standard tables²⁵ and gives the fourth equality subject to the condition $(n/2 + 1) > 0$ which is always satisfied since $n = 4, \dots, 8$. This result has exactly the same functional form as Eq. (12) of Ref. 19 if the substitution of the inverse G -matrix for the G -matrix suggested by Klippenstein immediately below his Eq. (12) is made. Differences between our result [Eq. (76)] and Klippenstein's Eq. (12) will be discussed elsewhere, along with connections of our canonical results to other work.

VI. SUMMARY

Canonical and microcanonical (energy-resolved) flexible transition state theory expressions for the rate constant have been derived for a variable reaction coordinate. The derivation is thorough and rigorous within the well-defined framework of physical approximations inherent in FTST. When the reaction coordinate is chosen to be the separation of centers-of-mass our treatment is an advantageous reformulation of the original Wardlaw–Marcus version of FTST, for an arbitrary reaction coordinate, it represents a significant extension of original FTST which was limited to a CMRC. The reformulation, extension to a VRC, and inclusion of the

Following Klippenstein,¹⁹ one can proceed to evaluate the above expression explicitly. The procedure parallels the canonical derivation in Sec. IV. One does the integral over p_{n+1} first. The remaining momentum integrals can be done analytically as can the integrals over the three external coordinates. The delta function integral over q_{n+1} is trivial, leaving a configuration integral over the internal (transitional mode) coordinates. The integrand contains the same determinant as in the canonical case [see Eq. (71)] but a different function of the transitional mode potential V_{tr} . We choose instead to obtain the microcanonical result by inverse Laplace transformation of $\mathcal{F}(R)$ because the derivation is shorter and the function to be transformed is well-known. Suppressing the R -dependence of both fluxes and using $\beta = (kT)^{-1}$ in place of T , we first rewrite the canonical flux as

$$\begin{aligned} \mathcal{F}(\beta) &= \frac{8\pi^2}{h} \left(\frac{2\pi}{h^2}\right)^{n/2} \int \cdots \int d\mathbf{q}_{\text{cm, int}} \\ &\times |\mathbf{A}_r^0 + \delta\mathbf{A}_r|^{1/2} \left(\frac{1}{\beta}\right)^{n/2+1} e^{-\beta V_{\text{tr}}}. \end{aligned} \quad (75)$$

The derivation consists of applying inverse Laplace transform to $\mathcal{F}(\beta)$ and identifying the result as $\mathbf{f}(\epsilon_{\text{tr}})$:

microcanonical case bring our version of FTST in line with related but different versions developed independently by Klippenstein^{8,19,26} and by Smith.^{4,27} The current approach facilitates comparisons to these other versions; detailed comparisons to the work of these authors as well as to the related work of Marcus²⁴ and Levine²⁸ will be provided in a subsequent paper. The present treatment replaces our previous work^{11,12} but results in an identical canonical rate expression for a CMRC and for a VRC. Our earlier treatments are, in hindsight, incomplete since they begin part way through the current full derivation and thereby implicitly involve some nontrivial assumptions.

The rate expressions apply to any definition of the separation distance between fragments in a barrierless recombination (or dissociation) that is held fixed during hindered rotations at the (flexible) transition state, and to any combination of fragment structure (atom, linear top, nonlinear top). The minimization of the rate constant with respect to this definition can be regarded as optimizing the reaction coordinate within a canonical or microcanonical framework. The expression is analytic except for a configuration integral whose evaluation generally requires numerical integration over internal angles (from one to five depending on the frag-

ments' structure). The primary components of the integrand are the transitional mode or hindering potential (V_{tr}) and the determinant of the inverse G -matrix associated with the external (overall) rotation and relative internal motion of the fragments. Only the general form of the integrand is given here; the specific form for each of the five possible combinations of fragment types is presented in a companion paper. The principal results are expressions for the canonical and microcanonical (E -resolved) rate constants. These are given below for the case of two nonlinear-top fragments; the systematic reduction to the simpler cases of nonlinear+linear top fragments, etc. (see Table I) is discussed in paper I for a CMRC and in paper IV for a VRC.

The canonical VRC-FTST rate constant expression is

$$k(T) = g_e g_\sigma^{-1} \frac{kT}{h} \frac{Q_{\text{cons}}^\dagger(T) Q_{tm}^\dagger(T)}{Q_{\text{react}}(T)}, \quad (77)$$

where

$$Q_{tm}^\dagger(T) = 8\pi^2 \left(\frac{2\pi}{\beta h^2} \right)^{n/2} \int \cdots \int d\mathbf{q}_{\text{cm,int}} |\mathbf{A}_r^0 + \delta\mathbf{A}_r|^{1/2} \times e^{-\beta \Delta V_{tr}(\mathbf{q}_{\text{cm,int}}, R^\dagger)} e^{-\beta V_{tr}^\dagger}. \quad (78)$$

The corresponding microcanonical expression is

$$k(E) = g_e g_\sigma^{-1} \frac{N^\dagger(E)}{h \rho_{\text{react}}(E)}, \quad (79)$$

where

$$N^\dagger(E) = \int_0^{E-V^\dagger} d\epsilon \rho_{\text{cons}}(\epsilon, R^\dagger) \frac{8\pi^2}{\Gamma\left(\frac{n}{2} + 1\right)} \left(\frac{2\pi}{h^2} \right)^{n/2} \times \int \cdots \int d\mathbf{q}_{\text{cm,int}} |\mathbf{A}_r^0 + \delta\mathbf{A}_r|^{1/2} \times (E - V^\dagger - \epsilon - \Delta V_{tr}(\mathbf{q}_{\text{cm,int}}, R^\dagger))^{n/2} \times u(E - V^\dagger - \epsilon - \Delta V_{tr}(\mathbf{q}_{\text{cm,int}}, R^\dagger)). \quad (80)$$

In Eqs. (78) and (80) the dependence of the determinant on the variables of integration and on R is implicit and $\Delta V_{tr} = V_{tr} - V^\dagger$ is the transitional mode potential evaluated with respect to $V^\dagger = V_{\text{MEP}}(\mathbf{r}_{i,\text{MEP}})$ [see text following Eq. (6)]. The determinant in Eqs. (78) and (80) can be expressed as

$$\begin{aligned} |\mathbf{A}_r^0 + \delta\mathbf{A}_r|^{1/2} &= \left\{ |\mathbf{A}_r^0| + \sum_{i=4}^n (\delta\mathbf{A}_r)_{ii} |(\mathbf{A}_r^0)^{ii}| + 2 \sum_{i=4}^{n-1} \sum_{j=i+1}^n (-1)^{i+j} (\delta\mathbf{A}_r)_{ij} |(\mathbf{A}_r^0)^{ij}| \right\}^{1/2} \\ &= |\mathbf{A}_r^0|^{1/2} \left\{ 1 + \sum_{i=4}^n \mu(\mathbf{b}\mathbf{b}^T)_{ii} \frac{|(\mathbf{A}_r^0)^{ii}|}{|\mathbf{A}_r^0|} + 2 \sum_{i=4}^{n-1} \sum_{j=i+1}^n (-1)^{i+j} \mu(\mathbf{b}\mathbf{b}^T)_{ij} \frac{|(\mathbf{A}_r^0)^{ij}|}{|\mathbf{A}_r^0|} \right\}^{1/2} \\ &= |\mathbf{A}_r^0|^{1/2} \left\{ 1 + \mu \left[\sum_{i=4}^n (\nabla F)_i^2 (G_r^0)_{ii} + 2 \sum_{i=4}^{n-1} \sum_{j=i+1}^n (\nabla F)_i (\nabla F)_j (G_r^0)_{ij} \right] \right\}^{1/2}. \end{aligned} \quad (81)$$

The first equality is Eq. (69); the second equality results from substituting for $\delta\mathbf{A}_r$ using Eqs. (68) and (47); and the last equality is obtained by recognizing that $(\nabla F^T)_i = (\nabla F)_i$ and that the coefficients of $(\nabla F)_i (\nabla F)_j$ are elements of the matrix $\mathbf{G}_r^0$ inverse to $\mathbf{A}_r^0$. Specifically,

$$(G_r^0)_{ij} \equiv (A_r^{0^{-1}})_{ij} = (-1)^{i+j} \frac{|(\mathbf{A}_r^0)^{ji}|}{|\mathbf{A}_r^0|}. \quad (82)$$

$\mathbf{G}_r^0$ is the reduced version of the kinetic energy matrix $\mathbf{G}^0$ obtained by removal of the last row and last column from the latter. $\mathbf{G}^0$ is the inverse of $\mathbf{A}^0$ and is defined by

$$2T = \mathbf{P}_{\text{cm}}^T \mathbf{G}^0 \mathbf{P}_{\text{cm}}, \quad (83)$$

where $\mathbf{P}_{\text{cm}} = \mathbf{A}^0 \dot{\mathbf{Q}}_{\text{cm}}$ and $\dot{\mathbf{Q}}_{\text{cm}}$ is defined in Eq. (11). Introducing the explicit form of $|\mathbf{A}_r^0|$ from paper I, yields

$$\begin{aligned} |\mathbf{A}_r^0 + \delta\mathbf{A}_r| &= (\mu F^2)^2 I_{1A} I_{1B} I_{1C} I_{2A} I_{2B} I_{2C} \sin^2 \theta_1 \sin^2 \theta_2 \\ &\times \left\{ 1 + \mu \left[\sum_{i=4}^n (\nabla F)_i^2 (G_r^0)_{ii} \right. \right. \\ &\quad \left. \left. + 2 \sum_{i=4}^{n-1} \sum_{j=i+1}^n (\nabla F)_i (\nabla F)_j (G_r^0)_{ij} \right] \right\}, \end{aligned} \quad (84)$$

where $F = F(\mathbf{q}_{\text{cm,int}}; R) = R_{\text{cm}}$ [Eq. (44)] has been introduced.

The form of the rate expressions and of the associated configuration integral have important conceptual and computational implications. These advantages are attributable to the analytical expression for the rate constant and the reduction in dimensionality of the configuration integral to an absolute minimum, i.e., there are no redundant variables of integration. Conceptual advantages are varied and include: factorization of the canonical rate for a CMRC into a gas collision rate multiplied by a steric factor which is the Boltzmann average of the transitional mode hindering potential; identification of a "flux correction factor" akin to that of Smith and which accounts for the change in flux through the DS when the RC is no longer the CM separation; analytical in-

vestigation of the limitations imposed by symmetry elements in the fragments on choices of reaction coordinate; analytical demonstration that the optimal RC at large separation or for a central potential is the CM separation. Detailed discussion of these and related features is deferred to future publications. The fact that the same variables of integration (Jacobi internal angles) are used for all reaction coordinate choices combined with analytic expressions for the integrand leads to advantages in the numerical evaluation of the configuration integral, namely ease of programming and reduced computation times. The treatment described in this paper has been included in the VariFlex program,²⁹ a general computer code for Variable reaction coordinate Flexible transition state theory written by a consortium including the present authors.

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APPENDIX A: PROOF THAT $\mathbf{a}_r^{-1} = \mathbf{g}_r$

Step 1: Express the conjugacy of $\mathbf{a}$ and $\mathbf{g}$ [see Eq. (51)] in terms of their matrix elements

$$\mathbf{a}\mathbf{g} = \mathbf{1} \Rightarrow \sum_{j=1}^{n+1} a_{ij}g_{jk} = \delta_{ik} \quad \text{for } i, k = 1, \dots, n+1. \quad (\text{A1})$$

Since $\mathbf{a}$ and $\mathbf{g}$ are symmetric, this condition can also be expressed

$$\sum_{j=1}^{n+1} a_{ij}g_{jk} = \sum_{j=1}^{n+1} a_{ji}g_{jk} = \delta_{ik}. \quad (\text{A2})$$

Step 2: Obtain expression for matrix elements of the product $\mathbf{a}_r \cdot \mathbf{g}_r$

$$\mathbf{a}_r \cdot \mathbf{g}_r = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & \ddots & & \vdots \\ \vdots & & & \\ a_{n1} & \cdots & & a_{nn} \end{pmatrix} \times \begin{pmatrix} g_{11} - \frac{g_{1,n+1}g_{n+1,1}}{g_{n+1,n+1}} & \cdots & g_{1n} - \frac{g_{1,n+1}g_{n+1,n}}{g_{n+1,n+1}} \\ \vdots & \ddots & \vdots \\ g_{n1} - \frac{g_{n,n+1}g_{n+1,1}}{g_{n+1,n+1}} & \cdots & g_{nn} - \frac{g_{n,n+1}g_{n+1,n}}{g_{n+1,n+1}} \end{pmatrix}. \quad (\text{A3})$$

The general expression for the (kl) th element of the product matrix is

$$(\mathbf{a}_r \mathbf{g}_r)_{kl} = \sum_{j=1}^n (a_r)_{kj}(g_r)_{jl}$$

$$\begin{aligned} &= \sum_{j=1}^n a_{kj}g_{jl} - \frac{g_{n+1,l}}{g_{n+1,n+1}} \sum_{j=1}^n a_{kj}g_{j,n+1} \\ &= \sum_{j=1}^{n+1} a_{kj}g_{jl} - a_{k,n+1}g_{n+1,l} - \frac{g_{n+1,l}}{g_{n+1,n+1}} \sum_{j=1}^n a_{kj}g_{j,n+1} \\ &= \delta_{kl} - \frac{g_{n+1,l}}{g_{n+1,n+1}} \left[a_{k,n+1}g_{n+1,n+1} + \sum_{j=1}^n a_{kj}g_{j,n+1} \right] \\ &= \delta_{kl} - \frac{g_{n+1,l}}{g_{n+1,n+1}} \sum_{j=1}^{n+1} a_{kj}g_{j,n+1} \\ &= \delta_{kl} - \delta_{k,n+1} \\ &= \delta_{kl} \end{aligned} \quad (\text{A4})$$

since $k = 1, 2, \dots, n$ for the *reduced* matrices, i.e., k never equals $n+1$. Thus $\mathbf{a}_r \cdot \mathbf{g}_r = \mathbf{1}$ and the proof is complete.

APPENDIX B: THE DETERMINANT OF $\mathbf{A}_r^0 + \delta\mathbf{A}_r$

For notational simplicity we derive the result with the substitution $(\mathbf{A}_r, \mathbf{A}_r^0, \delta\mathbf{A}_r) \rightarrow (\mathbf{A}, \mathbf{B}, \mathbf{C})$. Although in principle, knowledge of $\mathbf{B}$ and $\mathbf{C}$ is all that is required to evaluate the determinant of $\mathbf{A}$, in practice an algebraic formula for the determinant is likely to be cumbersome if evaluated in straightforward fashion, even with currently available algebraic evaluation software. However, the linearity property of determinants and the particular structures of the matrices $\mathbf{A}_r^0$ and $\delta\mathbf{A}_r$ leads to a dramatic simplification that permits a manageable closed form algebraic expression. The Appendix is organized as follows: Sec. 1 introduces convenient notation and lists relevant properties of the matrices $\mathbf{B}$ and $\mathbf{C}$; Sec. 2 derives a general expression for the determinant of a sum of two matrices; Sec. 3 contains the derivation of the determinant of $\mathbf{A}$ based on Secs. 1 and 2.

1. Preliminaries

Let $\mathbf{X}$ be a square matrix of order n . It is convenient to express $\mathbf{X}$ as a set of column vectors $\mathbf{x}_i$:

$$\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n], \quad \text{where } \mathbf{x}_i = \begin{bmatrix} X_{1i} \\ X_{2i} \\ \vdots \\ X_{ni} \end{bmatrix}. \quad (\text{B1})$$

In this notation, the determinant of $\mathbf{X}$ is written,

$$|\mathbf{X}| = |\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n|. \quad (\text{B2})$$

$\mathbf{B} = [\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n]$ and $\mathbf{C} = [\mathbf{c}_1, \mathbf{c}_2, \dots, \mathbf{c}_n]$ are square matrices of order n . The matrix $\mathbf{B}$ is symmetric, i.e., $b_{ij} = b_{ji}$. The matrix $\mathbf{C}$ can be written as the outer product $\mathbf{f}\mathbf{f}^T$ where $\mathbf{f}$ is a column vector with components $f_i = \mu^{1/2}(\nabla F)_i$ where $(\nabla F)_i$ is the gradient of the function F with respect to the i th component of the internal-external coordinates $\mathbf{q}_{\text{cm,int}}$:

$$\mathbf{f}_i = \begin{bmatrix} f_1 \\ f_2 \\ \vdots \\ f_n \end{bmatrix}. \quad (\text{B3})$$

The elements of $\mathbf{C}$ are of the form $C_{ij} = f_i f_j = f_j f_i = C_{ji}$ so that $\mathbf{C}$ is also symmetric. In the compact notation for matrices introduced above,

$$\mathbf{C} = [f_1 \mathbf{f}, f_2 \mathbf{f}, \dots, f_n \mathbf{f}]. \quad (\text{B4})$$

Since $f_i = 0$ for $i < 4$, the elements of $\mathbf{C}$ are zero everywhere except in a $(n-3) \times (n-3)$ block on the lower right diagonal.

2. General form of determinant for $\mathbf{A} = \mathbf{B} + \mathbf{C}$

In this subsection we illustrate the linearity property of determinants of matrices involved in a linear combination for 2×2 matrix and then employ this property for the $n \times n$ case. Specifically, it is shown that $|\mathbf{A}| = |\mathbf{B} + \mathbf{C}|$ can be expanded as a series of 2^n terms, each of which is the determinant of a matrix obtained by combining j columns of $\mathbf{B}$ and $(n-j)$ columns of $\mathbf{C}$, where $j = 0, 1, 2, \dots, n$.

If $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{C}$ are 2×2 then,

$$\begin{aligned} |\mathbf{A}| &= (B_{11} + C_{11})(B_{22} + C_{22}) - (B_{12} + C_{12})(B_{21} + C_{21}) \\ &= B_{11}B_{22} - B_{12}B_{21} + B_{11}C_{22} - C_{12}B_{21}C_{11}B_{22} - B_{12}C_{21} \\ &\quad + C_{11}C_{22} - C_{12}C_{21} \\ &= |\mathbf{b}_1, \mathbf{b}_2| + |\mathbf{b}_2, \mathbf{c}_2| + |\mathbf{c}_1, \mathbf{b}_2| + |\mathbf{c}_1, \mathbf{c}_2|. \end{aligned} \quad (\text{B5})$$

The last equality of Eq. (B5) demonstrates the linearity property of determinants for the $n=2$ case.

The general form the determinant of the sum of two $n \times n$ matrices is

$$|\mathbf{A}| = \sum |\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n|, \quad (\text{B6})$$

where each $\mathbf{x}$ is either $\mathbf{b}$ or $\mathbf{c}$. The sum extends over all possible substitutions such that $n_b = j$ and $n_c = n - j$ where j varies from 1 to n and n_b, n_c are the number of $\mathbf{x}$'s replaced by $\mathbf{b}$ or $\mathbf{c}$, respectively. Note that there are 2^n terms, each term contributing with a positive sign when the column vector indices are in numerical order, and that the first and last terms are, respectively, $\det(\mathbf{B})$ and $\det(\mathbf{C})$.

3. Specific form of determinant of $\mathbf{A}$ for $\mathbf{A} = \mathbf{B} + \mathbf{C}$

Expressions for $|\mathbf{A}|$ for $n=4$ and 8 (the smallest and largest values of n relevant in the rate constant expressions) are derived by consideration of the properties of $\mathbf{B}$ and $\mathbf{C}$ described in Sec. 1 and exploitation of several basic properties of determinants. These properties lead to enormous simplifications by greatly reducing the number of individual de-

terminant terms in the series expansion for $|\mathbf{A}|$ developed in Sec. 2, and by reducing the complexity of those terms which remain.

All terms in which two or more $\mathbf{x}$'s in an individual determinant term in Eq. (B6) are replaced by $\mathbf{c}$'s vanish. This is a result of the well known property that the determinant of a matrix with two or more identical columns (or rows) is zero and the particular form of $\mathbf{C}$ as revealed by Eq. (B4). Consider a particular such term from Eq. (B6), say with the first two $\mathbf{x}$'s replaced by $\mathbf{c}$'s and all remaining $\mathbf{x}$'s by $\mathbf{b}$'s:

$$\begin{aligned} |\mathbf{c}_1, \mathbf{c}_2, \mathbf{b}_3, \dots, \mathbf{b}_i, \dots, \mathbf{b}_n| &= |f_1 \mathbf{f}, f_2 \mathbf{f}, \mathbf{b}_3, \dots, \mathbf{b}_i, \dots, \mathbf{b}_n| \\ &= f_1 f_2 |\mathbf{f}, \mathbf{f}, \mathbf{b}_3, \dots, \mathbf{b}_i, \dots, \mathbf{b}_n| = 0. \end{aligned} \quad (\text{B7})$$

The first equality results from $\mathbf{c}_i = f_i \mathbf{f}$, (Sec. 1), and the second from the well-known property of matrices that multiplication of a column (or row) by a constant changes the determinant by the same constant factor. Clearly a value of zero is obtained by replacing any pair of $\mathbf{x}$'s by $\mathbf{c}$'s and this is equally true for any combination with three or more $\mathbf{x}$'s replaced $\mathbf{c}$'s. The nonzero terms are thus limited to the one with all $\mathbf{x}$'s replaced by $\mathbf{b}$'s [i.e., $\det(\mathbf{B})$] and to n with one $\mathbf{x}$ replaced by a $\mathbf{c}$ and all other $\mathbf{x}$'s replaced by $\mathbf{b}$'s, i.e.,

$$\begin{aligned} f_1 |\mathbf{f}, \mathbf{b}_2, \mathbf{b}_3, \dots, \mathbf{b}_i, \dots, \mathbf{b}_n|, \dots, f_i |\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3, \dots, \mathbf{f}, \dots, \mathbf{b}_n|, \dots, \\ f_n |\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3, \dots, \mathbf{b}_i, \dots, \mathbf{f}|. \end{aligned} \quad (\text{B8})$$

Since $f_i = 0$ for $i < 4$ (Sec. 1), only the last $n-3$ terms in the above sequence are nonzero.

Explicit expressions for $\det(\mathbf{A})$ for $n=4$ and 8 are developed below based on the above simplifications and on the symmetry of $\mathbf{B}$:

$n=4$:

$$\begin{aligned} |\mathbf{A}| &= |\mathbf{B}| + f_4 |\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3, \mathbf{f}| \\ &= |\mathbf{B}| + f_4 \begin{vmatrix} b_{11} & b_{12} & b_{13} & 0 \\ b_{21} & b_{22} & b_{23} & 0 \\ b_{31} & b_{32} & b_{33} & 0 \\ b_{41} & b_{42} & b_{43} & f_4 \end{vmatrix} \\ &= |\mathbf{B}| + (-1)^{4+4} f_4 f_4 |\mathbf{B}^{44}| \\ &= |\mathbf{B}| + C_{44} |\mathbf{B}^{44}|, \end{aligned} \quad (\text{B9})$$

where $|\mathbf{B}^{ij}|$ is the (i,j) th minor of the matrix formed from $\mathbf{B}$ by striking out the i th row and j th column and $C_{ij} = f_i f_j$:

$n=8$:

$$\begin{aligned} |\mathbf{A}| &= |\mathbf{B}| + f_4 |\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3, \mathbf{f}, \mathbf{b}_5, \mathbf{b}_6, \mathbf{b}_7, \mathbf{b}_8| + f_5 |\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3, \mathbf{b}_4, \mathbf{f}, \mathbf{b}_6, \mathbf{b}_7, \mathbf{b}_8| + f_6 |\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3, \mathbf{b}_4, \mathbf{b}_5, \mathbf{f}, \mathbf{b}_7, \mathbf{b}_8| \\ &\quad + f_7 |\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3, \mathbf{b}_4, \mathbf{b}_5, \mathbf{b}_6, \mathbf{f}, \mathbf{b}_8| + f_8 |\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3, \mathbf{b}_4, \mathbf{b}_5, \mathbf{b}_6, \mathbf{b}_7, \mathbf{f}| \\ &= |\mathbf{B}| + C_{44} |\mathbf{B}^{44}| + C_{55} |\mathbf{B}^{55}| + C_{66} |\mathbf{B}^{66}| + C_{77} |\mathbf{B}^{77}| + C_{88} |\mathbf{B}^{88}| + 2[(-1)^{4+5} C_{45} |\mathbf{B}^{45}| + (-1)^{4+6} C_{46} |\mathbf{B}^{46}| \\ &\quad + (-1)^{4+7} C_{47} |\mathbf{B}^{47}| + (-1)^{4+8} C_{48} |\mathbf{B}^{48}| + (-1)^{5+6} C_{56} |\mathbf{B}^{56}| + (-1)^{5+7} C_{57} |\mathbf{B}^{57}| + (-1)^{5+8} C_{58} |\mathbf{B}^{58}| \\ &\quad + (-1)^{6+7} C_{67} |\mathbf{B}^{67}| + (-1)^{6+8} C_{68} |\mathbf{B}^{68}| + (-1)^{7+8} C_{78} |\mathbf{B}^{78}|]. \end{aligned} \quad (\text{B10})$$

A general and compact expression for the determinant of $|\mathbf{A}|$ is:

$$|\mathbf{A}| = |\mathbf{B}| + \sum_{i=4}^n C_{ii} |\mathbf{B}^{ii}| + 2 \sum_{i=4}^{n-1} \sum_{j=i+1}^n (-1)^{i+j} C_{ij} |\mathbf{B}^{ij}|$$

$$= |\mathbf{B}| [1 + \sum_{i=4}^n C_{ii} (\mathbf{B}^{-1})_{ii} + 2 \sum_{i=4}^{n-1} \sum_{j=i+1}^n C_{ij} (\mathbf{B}^{-1})_{ij}]. \quad (\text{B11})$$

Equation (69) in the text is the version of Eq. (B11) with $(\mathbf{A}, \mathbf{B}, \mathbf{C})$ replaced by $(\mathbf{A}_r, \mathbf{A}_r^0, \delta \mathbf{A}_r)$, respectively.

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